

Preliminary Traversal of the Hodge Conjecture Problem Space

A registry-charted semantic traversal: CM-Weil classes, the K3 /
Kuga-Satake regime, the three-shape frontier, and the scoped status of the
transcendence question

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Contents

Abstract	1
Posture and scope	2
1. Traversal protocol	2
2. Objects and definitions	3
3. Boundary nodes	4
4. Node 0: abelian threefold $E_1 \times E_2 \times E_3$	5
5. Node 1: Weil fourfold with CM by $\mathbb{Q}(\sqrt{-3})$	5
6. Node 2: Weil fourfold with CM by $\mathbb{Q}(\sqrt{-7})$	7
7. Node 3: K3 surface in degree two	7
8. Node 4: Kuga-Satake weld	8
9. Node 5: other CM types and scope widening	9
10. Register closure and the three-shape frontier	10
11. Weld criteria	10
12. What the traversal establishes	11
13. Sources and standing	11
14. Conclusive scoped status	12
15. Outside-frame traversal and the transcendence question	13
16. Final scope guard	14
References and source anchors	15

Abstract

This document presents a semantic traversal of selected Hodge-theoretic problem spaces. It does not claim to prove, refute, or decide the Hodge conjecture. Its purpose is more limited and more precise: to define an explicit family of objects, bridges, predicates, and stopping conditions; to push each bridge until it either closes or produces a definable residue; and to record the resulting frontier without inflating bridge failure into a universal statement about all possible mathematics.

The traversal distinguishes co-location from weld. A registry may locate two structures at a common address or in an adjacent conceptual register, but such co-location does not transfer truth. Transfer requires a weld: a theorem or construction that rigidly couples the frames so that information in one frame forces information in the other. This distinction controls the entire analysis.

The result is a three-shape boundary for the portion of the Hodge problem space examined here. Divisor-level cases, such as degree-two Hodge classes on K3 surfaces, close by Lefschetz (1,1). CM anti-invariant Weil-type classes expose a no-weld regime: norm-type reattachments land in the conjugation-invariant component, while spectral or harmonic data certify the wrong predicate. Kuga-Satake constructions provide a third regime in which a genuine Hodge-structure weld exists, but the algebraicity of the corresponding cycle remains open in general. A final outside-frame analysis examines absolute Hodge theory and period-transcendence methods. These outside frames do not establish Hodge-non-algebraicity; the genuine Lindemann-style transcendence machinery concerns periods, not rational Hodge classes as cycle classes. The traversal therefore earns only a scoped conclusion: within the defined bridge family and the tested outside frames, the transcendence reading does not cross the Hodge predicate, while the algebraicity question remains open.

Posture and scope

This is a semantic traversal, not a proof attempt. The goal is to define the problem space of the Hodge conjecture in decidable language, push available bridges until each either closes or generates a definable residue, and log the result so that the conclusion emerges from the completed description rather than from assertion.

No claim is made or implied about the truth, falsity, or provability of the Hodge conjecture. Where a bridge stops, the stop is a statement about that bridge's reach within its stated frame, not about the existence or non-existence of algebraic cycles. The Hopf registry appears here only as a declared chart: a coordinate system for locating and comparing frames. It does not function as a generative mechanism, a truth oracle, or a substitute for a theorem.

The following tier labels govern the document.

Tier I — Verifiable. Standard, checkable mathematics, stated as fact.

Tier II — Structural. Exact correspondences and classifications read within the framework after the Tier-I objects have been fixed.

Tier III — Programmatic. Forward-looking claims, conjectural directions, and author-defined interpretive structures. These are not asserted as established mathematics.

1. Traversal protocol

The traversal follows a fixed protocol so that termination is a design property rather than an aspiration.

1. **Initial condition.** Each node begins from an explicitly stated mathematical object and target predicate.
2. **Finite reduction.** From the stated initial condition, admissible reductions proceed finitely unless the reduction generates an irreducible residue.
3. **Residue as definable stop.** An irreducible residue is a point at which a reduction cannot continue inside the stated frame. When that point is detectable and nameable, it becomes a recorded stopping condition rather than an unbounded recursion.
4. **Two stop types.** An **exhaustion stop** occurs when every condition at the node reduces to decidable language; the branch closes and its verdict is recorded. An **irreducible-residue stop** occurs when the stated frame cannot carry the reduction further; the residue's coordinate and type are recorded, the node is marked open relative to that frame, and the traversal departs.
5. **Depart and do not re-enter.** A logged node is not revisited unless a new bridge is explicitly added to the scope. Ground already exhausted is not re-covered.

This mirrors the Dynamical Semantic Traversal cycle: reduce until marginal structural return is non-positive, record the residue with its coordinate, and depart.

The word *irreducible* always means irreducible relative to the stated bridge, frame, and predicate. It does not mean intrinsically irreducible relative to all possible mathematics.

2. Objects and definitions

2.1 The Hodge conjecture

Let X be a smooth projective variety over $\mathbb{C}$. For each k , the rational Hodge classes are

$$\text{Hdg}^k(X) = H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X).$$

An algebraic cycle of codimension k is a finite $\mathbb{Q}$ -combination of codimension- k sub-varieties. Its cohomology class lies in $\text{Hdg}^k(X)$. The Hodge conjecture asserts the reverse inclusion: every rational Hodge class is the class of an algebraic cycle.

2.2 Residue lens

For $\alpha \in \text{Hdg}^k(X)$ define the residue notation

$$\mathcal{I}(\alpha) = \text{obstruction of } \alpha \text{ to lying in the } \mathbb{Q}\text{-span of algebraic-cycle classes.}$$

This notation does not solve the Hodge question. It records where the open predicate sits. One has

$$\mathcal{I}(\alpha) = 0 \iff \alpha \text{ is algebraic.}$$

Thus deciding $\mathcal{I}(\alpha) = 0$ is logically identical to deciding the Hodge question for α . The residue lens is bookkeeping, not a reduction that removes the difficulty.

2.3 Registry as chart

A Hopf configuration

$$S^1 \hookrightarrow S^{2n+1} \rightarrow \mathbb{CP}^n$$

is used as a declared scaffold for an extendable family of circumscribing shells. Its role here is strictly chart-like: it assigns frames such as algebraic, spectral, and arithmetic frames stable addresses and makes adjacency between frames statable.

By the registry/map separation, a descriptively complete registry is inert as a truth oracle. A chart locates and relates; it does not decide. Two notions remain distinct throughout:

- **Co-location:** two frames sit at the same chart address or in a statable adjacency relation.
- **Weld:** a theorem, construction, or correspondence rigidly coupling two frames so that a fact in one frame forces a fact in the other.

Co-location may be necessary for a weld, but it is not sufficient.

3. Boundary nodes

3.1 Projective space: $\mathbb{CP}^n$

Tier I. The cohomology ring of complex projective space is

$$H^*(\mathbb{CP}^n) = \mathbb{Z}[\omega]/(\omega^{n+1}),$$

with ω the hyperplane class. Every cohomology class is an integer multiple of a power ω^k , and ω^k is the class of a linear subspace $\mathbb{CP}^{n-k}$. Hence the Hodge conjecture holds on $\mathbb{CP}^n$ trivially.

Stop. Exhaustion. Matching spectral or registry data to powers of ω engages none of the conjecture's difficult content. Projective space is a closed boundary node, not the frontier.

3.2 The Hopf total space as category constraint

Tier I. The sphere S^{2n+1} has odd real dimension and rational cohomology only in degrees 0 and $2n + 1$. It is not a complex projective variety and does not carry a Hodge structure of the type required for the Hodge conjecture. The map

$$S^{2n+1} \rightarrow \mathbb{CP}^n$$

is a smooth $U(1)$ -bundle, not a morphism of projective varieties.

Consequence. A literal instruction to lift an algebraic class to the Hopf total space has no referent in the category where algebraic cycles live. This does not prevent the chart use of the Hopf configuration. Co-location remains statable. It prevents only the stronger reading in which the shell itself functions as an algebraic carrier.

4. Node 0: abelian threefold $E_1 \times E_2 \times E_3$

Initial condition. Let $A = E_1 \times E_2 \times E_3$ be a product of three elliptic curves. Then $\dim_{\mathbb{C}} A = 3$. The target is the degree-four Hodge classes.

Tier I. The Hodge numbers of a product of three elliptic curves satisfy

$$h^{p,q} = \binom{3}{p} \binom{3}{q}.$$

Thus $h^{2,2} = \binom{3}{2}^2 = 9$, with Hodge diamond

$$\begin{array}{ccccc} & & 1 & & \\ & 3 & & 3 & \\ & 3 & 9 & 3 & \\ 1 & 9 & & 9 & 1 \\ & 3 & 9 & 3 & \\ & 3 & & 3 & \\ & & 1 & & \end{array}$$

For a generic product with no extra isogenies or complex multiplication, the rational Hodge classes are generated by divisor classes and are therefore algebraic. The genuinely open non-divisor Hodge classes do not appear on a generic threefold of this type.

Stop. Exhaustion. The node is closed and departed.

Frontier relocation. The canonical home of non-divisor Weil classes lies in abelian fourfolds with complex multiplication by an imaginary quadratic field, as in Weil's construction. The Hodge conjecture for these classes remains open in general, with scattered special cases resolved.

Tier II chart note. An imaginary quadratic CM field $\mathbb{Q}(\sqrt{-d})$ occupies the negative-discriminant register of the same quadratic-field structure that appears in Gaussian-period factorizations. For primes $p \equiv 3 \pmod{4}$, the sign convention $p^* = (-1)^{(p-1)/2}p < 0$ gives the negative register. This is recorded as co-location, not as a weld.

5. Node 1: Weil fourfold with CM by $\mathbb{Q}(\sqrt{-3})$

5.1 Initial condition and target

Let A be an abelian fourfold of Weil type with complex multiplication by

$$K = \mathbb{Q}(\sqrt{-3}).$$

Then $\dim_{\mathbb{C}} A = 4$, and the Hodge numbers satisfy

$$h^{p,q} = \binom{4}{p} \binom{4}{q}.$$

The middle values include

$$h^{2,0} = 6, \quad h^{2,1} = 24, \quad h^{2,2} = \binom{4}{2}^2 = 36.$$

Inside $H^{2,2}(A)$ sits a distinguished subspace $W_A \subset \text{Hdg}^2(A)$ of Weil classes. These classes are characterized as a Galois-anti-invariant eigenspace of the K -action on cohomology and are not contained in the subring generated by divisor classes. The target is precisely this non-divisor component.

5.2 Bridge A: norm-cycle reattachment

The programmatic algebraic reattachment considered here uses a norm

$$N(g) = g\bar{g},$$

the product of an element with its Galois conjugate over K . The norm is, by construction, invariant under $\text{Gal}(K/\mathbb{Q})$. Hence it produces classes in the Galois-fixed component.

The target Weil classes occupy the Galois-anti-invariant component. The norm therefore lands in the wrong eigenspace. It cannot produce the target classes by this construction.

Stop. Irreducible residue relative to Bridge A.

- **Type:** frame mismatch.
- **Coordinate:** the Galois-anti-invariant eigenspace of the K -action on $H^{2,2}(A)$.
- **Missing weld:** a reattachment operation preserving the anti-invariant component and realized by an algebraic cycle.

This records only the reach of the norm bridge. It does not assert that no algebraic cycle exists in W_A .

5.3 Bridge B: Beltrami / Dirac spectral bridge

The abelian variety A is compact Kähler. By Hodge theory, every class in $H^{2,2}(A, \mathbb{Q})$ has a unique harmonic representative. Laplace-Beltrami and Dirac-type spectral data can certify harmonicity. However, harmonicity is automatic for Hodge classes. The open predicate is algebraicity, not harmonicity.

Thus spectral membership carries no information that separates algebraic Hodge classes from non-algebraic Hodge classes within the Hodge locus. The spectral bridge resolves the wrong predicate.

Stop. Irreducible residue relative to Bridge B.

- **Type:** predicate mismatch.
- **Coordinate:** the gap between harmonic representative and algebraic cycle.
- **Missing weld:** a theorem coupling a spectral invariant to the cycle-class predicate.

5.4 Node status

Both bridges stop at the same seam from opposite directions. Algebraicity is a Galois-rational cycle-class property. The norm bridge reaches the conjugation-invariant component. The spectral bridge certifies harmonicity, which all Hodge classes already possess. Neither bridge reaches the Galois-anti-invariant Weil component as an algebraic cycle.

Node status. Departed with two logged residues.

6. Node 2: Weil fourfold with CM by $\mathbb{Q}(\sqrt{-7})$

This node tests whether the residue types from Node 1 depend on the specific CM field.

Let A be an abelian fourfold of Weil type with CM by

$$K = \mathbb{Q}(\sqrt{-7}).$$

The field has class number one and occupies the negative register associated with the $p = 7$ Gaussian-period atom.

For Bridge A, the norm $N(g) = g\bar{g}$ remains invariant under $\text{Gal}(K/\mathbb{Q})$ for every imaginary quadratic field. Weil classes remain conjugation-anti-invariant. Therefore the same frame-mismatch residue recurs, with only the field coordinate changed.

For Bridge B, compact Kähler geometry again supplies harmonic representatives for all Hodge classes. Spectral data certify harmonicity, not algebraicity. The same predicate-mismatch residue recurs.

Result. The prediction is confirmed. The residue types do not depend on the discriminant d ; the field changes the coordinate, not the residue mechanism.

7. Node 3: K3 surface in degree two

7.1 Initial condition

Let X be a K3 surface. The target is the degree-two Hodge classes.

Tier I. The second integral cohomology lattice is

$$H^2(X, \mathbb{Z}) \cong U^3 \oplus E_8(-1)^2,$$

with signature $(3, 19)$. The Hodge numbers are

$$(h^{2,0}, h^{1,1}, h^{0,2}) = (1, 20, 1).$$

The Hodge classes in H^2 are the Néron-Severi group $NS(X)$, with Picard rank $1 \leq \rho \leq 20$. The transcendental lattice is

$$T(X) = NS(X)^\perp,$$

of rank $22 - \rho$, and it carries the $(2, 0)$ part.

7.2 Closure by Lefschetz (1,1)

The node tests whether the residue types from the Weil/CM regime are universal. They are not. By the Lefschetz $(1, 1)$ theorem, every integral $(1, 1)$ class on a compact Kähler manifold is the class of a divisor. Therefore every degree-two Hodge class on a K3 surface is algebraic.

Stop. Exhaustion by theorem.

Where a genuine weld exists - Lefschetz $(1, 1)$ coupling $(1, 1)$ cohomology directly to divisor classes - the prior failure modes disappear. The CM residues are not universal. They belong to middle-codimension, non-divisor classes in a conjugation-anti-invariant direction, not to transcendental classes in general.

8. Node 4: Kuga-Satake weld

8.1 Hodge-structure weld

Moving from K3 surfaces in degree two toward open middle-codimension problems, such as classes on $K3 \times K3$, leads to the Kuga-Satake construction. For a polarized weight-two Hodge structure of K3 type with $h^{2,0} = 1$, the even Clifford algebra $C^+(T(X), q)$ of the transcendental lattice yields an abelian variety A_{KS} and an embedding of Hodge structures

$$H^2(X)_{\text{prim}} \hookrightarrow H^1(A_{KS}) \otimes H^1(A_{KS}).$$

This is a genuine weld at the level of Hodge structures. It does more than co-locate the K3 and abelian registers; it couples them by a theorem.

8.2 Algebraicity of the weld

A morphism of Hodge structures does not by itself produce an algebraic cycle realizing that morphism. The algebraicity of the Kuga-Satake correspondence is known in special cases and remains open in general.

Stop. Irreducible residue of a third type.

- **Type:** Hodge-level weld exists; algebraicity of the weld remains open.
- **Coordinate:** the algebraicity of the Kuga-Satake correspondence class in $H^*(X \times A_{\text{KS}})$.

This is neither the frame mismatch of the norm bridge nor the predicate mismatch of the spectral bridge. The frames do couple, but the cycle realizing the coupling remains unproduced in general.

8.3 Structural consequence

The one route that makes K3 transcendental data tractable runs toward the abelian register. It provides a Hodge-structure weld, but the required algebraic realization remains the open predicate.

9. Node 5: other CM types and scope widening

This node tests whether the imaginary quadratic case was load-bearing or incidental.

9.1 Real multiplication

A totally real field is not a CM field. Abelian varieties with real multiplication do not carry Weil-type exotic classes of this construction because the phenomenon requires a complex conjugation acting nontrivially across $H^{1,0}$ and $H^{0,1}$.

Finding. The target is absent. The node is vacuous for the residue.

9.2 Higher-degree CM fields

Let K be a CM field with totally real subfield K_0 and $[K : \mathbb{Q}] = 2e$. Nondegenerate CM types produce exotic Hodge classes. The relative norm

$$N_{K/K_0}$$

is K_0 -valued and fixed by the complex conjugation of K/K_0 . Exotic classes occupy the conjugation-anti-invariant component. The norm therefore misses them by the same mechanism as in the imaginary quadratic case.

Finding. The frame-mismatch residue recurs, now with conjugation living in K/K_0 .

9.3 Higher-dimensional Weil type

The Weil-type construction generalizes to abelian $2n$ -folds. The same residues recur: conjugation-invariant reattachment misses the anti-invariant target, and spectral data certify harmonicity rather than algebraicity. Dimension changes the coordinate, not the residue type.

9.4 Corrected scope

The load-bearing structure is not the specific imaginary quadratic field and not the dimension. It is the presence of a CM conjugation pair K/K_0 whose anti-invariant eigenspace hosts the target, against a reattachment operation that is conjugation-invariant by construction.

10. Register closure and the three-shape frontier

The verified nodes produce a graded boundary rather than a single wall.

Regime	Locus	Weld status	Residue
Divisor / $(1, 1)$	K3 degree two; divisor classes	Weld exists and is algebraic by Lefschetz $(1, 1)$	None; closed
CM anti-invariant	Weil-type classes, CM pair K/K_0	No weld from the defined bridges; norm is conjugation-fixed and spectrum certifies harmonicity	Frame mismatch and predicate mismatch
Kuga-Satake	K3 transcendental data to abelian H^1	Weld exists at Hodge-structure level	Algebraicity of the weld remains open

Widened closure, scoped exactly. Wherever the target Hodge classes are the conjugation-anti-invariant eigenspace of a CM action, the norm-cycle bridge terminates in a frame-mismatch residue and the spectral bridge terminates in a predicate-mismatch residue. The field and dimension enter as residue coordinates. Where no CM target exists, the register is vacuous. Where a Hodge-structure weld exists, the remaining residue is the algebraicity of that weld.

This statement concerns the defined bridges and their scoped reach. It does not assert that any class is non-algebraic, that the Hodge conjecture holds or fails, or that no other method can reach the target.

11. Weld criteria

The traversal identifies the open node by shape rather than by gesture.

In the Kuga-Satake regime, a crossing weld would require a construction exhibiting the Kuga-Satake correspondence

$$H^2(X)_{\text{prim}} \hookrightarrow H^1(A_{\text{KS}})^{\otimes 2}$$

as the class of an algebraic cycle on $X \times A_{\text{KS}}$ for general X , not merely as a morphism of Hodge structures.

In the CM regime, a crossing weld would require a reattachment operation that preserves the conjugation-anti-invariant component and is realized by an algebraic cycle. The field norm does not have this property because it is conjugation-invariant by construction.

No such construction is asserted to exist in either regime. The criterion states what would have to be produced.

12. What the traversal establishes

The traversal establishes the following scoped results.

1. The examined Hodge frontier has a three-shape graded boundary: closed divisor cases, no-weld CM anti-invariant cases under the defined bridges, and Hodge-level weld cases whose algebraicity remains open.
2. In the CM regime, two distinct and definable failure modes appear: frame mismatch for norm-type reattachment and predicate mismatch for spectral-harmonic bridges.
3. The load-bearing structure in the CM regime is the conjugation pair K/K_0 , not a particular field or dimension.
4. Projective space and generic elliptic threefold products close as boundary nodes.
5. K3 surfaces in degree two close by Lefschetz $(1, 1)$.
6. Kuga-Satake links K3 transcendental data to the abelian register by a genuine Hodge-structure weld while leaving the algebraicity of the correspondence open in general.
7. Concrete weld criteria can be stated for both the CM and Kuga-Satake regimes.

The traversal does not establish any of the following.

1. The truth, falsity, or formal independence of the Hodge conjecture.
2. The algebraicity or non-algebraicity of any open target class.
3. The non-existence of bridges outside those defined and tested here.
4. An intrinsic impossibility theorem about all methods.

The contribution is the protocol, the explicit residue coordinates and types, the three-shape typology, the scoped closures, and the strict separation of co-location from weld and of bridge reach from cycle existence.

13. Sources and standing

The Tier-I mathematical material draws on standard results: the cohomology ring of projective space; Hodge numbers of abelian varieties; Hodge theory on compact

Kähler manifolds; the Lefschetz $(1, 1)$ theorem; the K3 lattice and the Néron-Severi/transcendental decomposition; Weil’s construction of exotic Hodge classes on fourfolds of Weil type; partial results of Schoen and van Geemen; the theory of complex multiplication and CM types, including work of Hazama and Murty; the Kuga-Satake construction of Kuga, Satake, and Deligne; and known algebraicity cases for Kuga-Satake correspondences, including work of Morrison and Paranjape.

The registry/map separation invoked here belongs to the companion program and functions only as an organizing chart. It does not replace standard mathematics. The Tier-III elements remain author-defined structural language rather than established external terminology.

14. Conclusive scoped status

14.1 Verdict within the posed instrument

Within the question as framed and the answer-space as probed - the Hopf configuration read as an infinite registry chart, together with the norm-arithmetic and Beltrami/Dirac spectral bridges - neither the Hodge conjecture nor its negation is supported. Each bridge terminates in a definable residue before producing or excluding a cycle.

The proper verdict is therefore:

Undecidable within the problem space and bridge family as posed.

This is a verdict about the question/answer-space pairing. It is not a verdict about the absolute logical status of the Hodge conjecture.

14.2 Earning condition

The scoped verdict is earned when continued reframing inside the instrument yields semantic nuance but no structural gain toward producing or excluding an algebraic cycle.

The tested reframings inside the instrument include:

- changing the CM field from -3 to -7 ;
- changing from imaginary quadratic to general CM conjugation pairs;
- changing from abelian fourfolds to higher-dimensional Weil type;
- replacing norm language with any conjugation-invariant reattachment;
- replacing Beltrami language with Dirac or general spectral language.

Each of these reframings relabels the coordinate but preserves the residue type.

Reframings that do produce structural gain do so by changing carrier or scope. K3 degree-two classes close by Lefschetz $(1, 1)$. Kuga-Satake produces a new Hodge-structure weld but relocates the residue to algebraicity of the correspondence. These gains map the frontier’s shape without crossing it.

14.3 Refused upgrades

Two stronger upgrades are explicitly refused.

Upgrade 2: intrinsic Lambda-irreducibility. Not earned. Failure of the defined instrument is not a survey of all possible reductions. The Kuga-Satake weld already shows that the mathematical space outside the initial two bridges contains additional structure.

Upgrade 3: transcendence or refutation from outside. Not earned by the original traversal. A transcendence-of- π -style proof would require a welded outside structure that converts algebraic-cycle failure into a contradiction. Such a proof would refute the Hodge conjecture for the exhibited class. The original traversal locates no weld in that transcendence-ward direction. The CM anti-invariant directions remain unwelded under the tested bridges, while the Kuga-Satake weld points toward algebraicity rather than away from it.

14.4 Durable yield

The durable result is cartographic. The traversal converts a vague frontier into a located three-shape boundary: closed divisor cases, no-weld CM anti-invariant cases under the defined bridges, and Hodge-weld-open Kuga-Satake cases. It demonstrates the chart-not-oracle character of the registry: a registry can locate the frontier, but a decision requires a weld.

15. Outside-frame traversal and the transcendence question

The previous section refused a transcendence upgrade from the original bridge family. This section records the corresponding outside-frame traversal. It tests the genuine available circumscribing frames rather than leaving the outside gesture unexamined.

15.1 Node 6: Deligne absolute Hodge theory

Deligne's theorem that Hodge classes on abelian varieties are absolutely Hodge supplies a real outside frame. The frame involves comparison across de Rham, Betti, and étale realizations and the action of field automorphisms. Algebraic cycle classes are absolutely Hodge, so absolute Hodge behavior supplies a necessary condition for algebraicity.

The Weil/CM Hodge classes pass this condition. Thus the outside frame welds the target classes to a property algebraic cycles must have, without producing the algebraic cycle itself.

Stop. Hodge/absolute-Hodge weld exists; cycle realization remains open.

This frame leans toward algebraicity in the limited sense that the target classes satisfy a necessary algebraicity-compatible condition. It provides no weld toward Hodge-non-algebraicity.

15.2 Node 7: functional and period transcendence

The genuine Lindemann-style outside theory in this region is the functional transcendence theory of period maps, including Ax-Schanuel-type results for variations of Hodge structure and related o-minimal/Pila-Zannier methods. This is the modern mathematical home of analytic transcendence in Hodge-theoretic settings.

However, the predicate differs. Period transcendence concerns the transcendence of period coordinates, such as entries of period matrices or functional relations among period maps. The Hodge conjecture concerns whether a rational cohomology class

$$\alpha \in H^{2k}(X, \mathbb{Q}) \cap H^{k,k}(X)$$

lies in the $\mathbb{Q}$ -span of algebraic-cycle classes. Such an α is a rational cohomology class. Its possible failure to be algebraic is not number-transcendence in the Lindemann sense.

Stop. Predicate mismatch.

The actual transcendence theory addresses periods, not the cycle-algebraicity of rational Hodge classes. Therefore the Lindemann analogy does not transfer to a Hodge-non-algebraicity verdict.

15.3 Node 8: motivic and standard-conjectural sweep

Other major outside frameworks - motives, the Tate conjecture, standard conjectures, and absolute-Hodge methods - bear on the Hodge problem by predicting or organizing algebraicity, not by producing a theorem of Hodge-non-algebraicity. No known outside frame supplies a welded path from the examined target classes to a non-algebraicity conclusion.

Stop. No transcendence-ward weld found.

15.4 Outside-frame conclusion

The transcendence avenue is not established. The strongest available analytic transcendence machinery concerns a different predicate. The outside frames that do weld the target classes to additional structure lean algebraicity-compatible rather than transcendence-ward: Deligne's absolute Hodge theorem and Kuga-Satake both supply structure that algebraic cycles would satisfy, while leaving cycle production open.

This does not prove algebraicity. Passing necessary conditions is not sufficient. The Hodge conjecture remains open. The outside traversal establishes only that the Lindemann-style transcendence reading fails to cross the Hodge predicate as posed.

16. Final scope guard

The phrase **undecidable within the problem space as posed** must not be compressed to **undecidable**. The logical status of the Hodge conjecture is open and

untouched by this traversal. Instruments outside the defined bridge family remain unconstrained unless explicitly tested. The outside-frame supplement tests several major available frames and records their stops, but it does not survey all possible future mathematics.

The traversal's earned conclusion is exactly this: within the defined registry, bridge family, and tested outside frames, every path either closes, relocates the frontier, or terminates in a definable residue before producing or excluding an algebraic cycle. The resulting map is coherent, scoped, and mathematically accountable; it is not a proof or refutation of the Hodge conjecture.

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